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Micro-aeration enhances lactic acid production in food waste fermentation

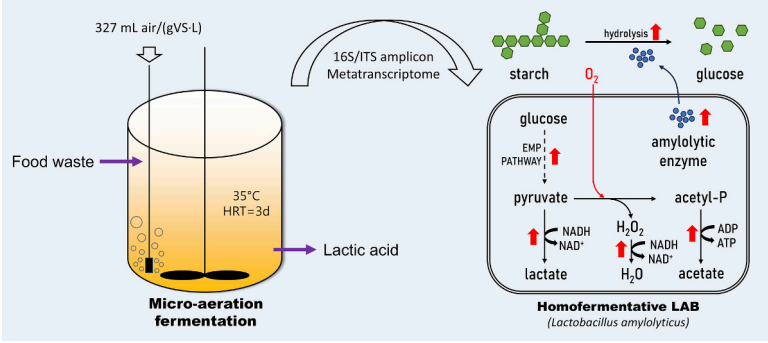
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HIGHLIGHTS

- Low-to-moderate micro-aeration significantly enhances lactic acid fermentation of food waste.
- An aeration rate of 327 mL air/(gVS·L) achieved the highest steady-state lactic acid concentration.
- Micro-aeration promoted the activity of homofermentative *Lactobacillus amylolyticus*.
- Fungal oxidative phosphorylation and TCA cycle activity were stimulated under micro-aeration.
- Heterofermentative LAB lost their competitive advantage under micro-aerobic conditions.

GRAPHICAL ABSTRACT



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ABSTRACT

Micro-aeration is a promising strategy to enhance hydrolysis and acidogenesis in anaerobic fermentation systems, but its role in lactic acid (LA) fermentation of food waste (FW) remains unclear. Here, four 1.5-L semi-continuous reactors (35 °C, HRT = 3 d) were run at dissolved oxygen < 0.5 mg/L with aeration loads of 0, 109, 327 and 872 mL air/(gVS·L). Moderate aeration (327 mL air/(gVS·L)) increased steady-state LA to 17.68 ± 1.14 g/L (0.40 g/gVS), 35.5 % higher than strict anaerobiosis, with LA contributing > 85 % of product COD and maximum hydrolysis and acidification efficiencies. In contrast, excessive aeration (872 mL air/(gVS·L)) decreased LA yield. Amplicon and metatranscriptomic analyses showed that micro-aeration enriched homofermentative *Lactobacillus amylolyticus* and activated pyruvate oxidase–acetyl phosphate–acetate and NADH peroxidase pathways, enhancing redox balancing and starch hydrolysis. Overall, controlled micro-aeration steered metabolism toward energy-efficient homofermentative LAB, improving LA yield and process stability.

1. Introduction

Anaerobic biological treatment is a widely adopted strategy for managing organic solid waste. In the absence of oxygen, anaerobic microorganisms degrade organic matter into low-molecular-weight products to obtain energy for their metabolism and growth (Zhang et al., 2014; Zhou et al., 2018). These biochemical processes allow for the valorization of organic waste into a range of high-value products, such as methane (CH_4), hydrogen (H_2), organic acids, etc., making this approach particularly attractive for resource recovery from a wide range of substrates, including agricultural residues, food waste (FW), and herbal-medicine by-products. (Prasanna Kumar et al., 2024; Wang et al.,

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2020).

Among various anaerobic treatment technologies, anaerobic digestion (AD) is the most widely implemented due to its high conversion efficiency and energy recovery potential. It is governed by four successive and tightly interlinked microbial steps: hydrolysis, acidogenesis, acetogenesis, and methanogenesis (Ren et al., 2018). Traditionally, AD is regarded as requiring strict anaerobic operating conditions, mainly owing to the high oxygen sensitivity of methanogens. Specifically, the key metabolic enzymes of these organisms, such as methyl-coenzyme M reductase, harbor low-redox-potential metal cofactors (e.g., Ni and Fe) that are highly susceptible to oxidative inactivation (Thauer, 2019). Moreover, intracellular oxygen can generate reactive oxygen species (ROS), such as superoxide (O_2^-) and hydrogen peroxide (H_2O_2), leading to secondary oxidative damage (Lu and Imlay, 2021). Beyond methanogens, syntrophic bacteria responsible for converting longer-chain volatile fatty acids into acetic acid, H_2, and CO_2 have also been reported to be highly sensitive to oxygen (Magalhães et al., 2025). Theoretically, even trace amounts of O_2 can disrupt the low hydrogen partial pressures essential for syntrophic metabolism and inhibit the activity of key hydrogenases (Sieber et al., 2012). Collectively, based on traditional theories, the presence of oxygen in AD systems can significantly reduce microbial activity, impair the acetogenic and methanogenic functions of the microbial community, and ultimately lead to decreased methane yields.

However, in recent years, a growing body of research has demonstrated that micro-aeration can enhance the performance of AD. Among the four stages of AD, micro-aeration can stimulate hydrolysis and acidogenesis. Specifically, it can activate hydrolytic bacterial communities to produce higher levels of extracellular enzymes such as amylases, proteases, and cellulases, thereby accelerating the breakdown of macromolecular substrates (Johansen and Bakke, 2006; Sawatdeenarunat et al., 2017). This enzymatic enhancement increases the availability of monomers for fermentative bacteria, facilitating the production of volatile fatty acids (VFAs) and other fermentation products (Yin et al., 2016). Acetogenesis is thermodynamically unfavorable. If its progress is slowed or inhibited, intermediate short-chain fatty acids (SCFAs), particularly propionic acid and butyric acid, may accumulate (Han et al., 2020). Trace amounts of oxygen (O_2) have been shown to stimulate the conversion of SCFAs into acetic acid, thereby increasing the availability of substrates for methanogens (Lim and Wang, 2013). Furthermore, under controlled aeration rates, facultative bacteria can directly utilize O_2 to oxidize VFAs into CO_2, thereby mitigating VFA accumulation and alleviating pH-induced inhibition of methanogenic archaea (Nguyen et al., 2019). Importantly, oxygen introduced through micro-aeration is rapidly consumed by facultative microorganisms, thereby maintaining an anaerobic environment suitable for methanogenesis (Nguyen and Khanal, 2018). As a result, micro-aeration does not inhibit the AD process; on the contrary, it can promote methanogenic activity, improve system stability, and enhance methane yields (Jenicek et al., 2014; Li et al., 2024). Given its ability to stimulate hydrolysis and acidogenesis, micro-aeration holds potential not only for enhancing biogas production but also for application in anaerobic fermentation systems where high-value organic acids are the desired end products.

In recent years, this concept has been extended from biogas-oriented AD to acidogenic fermentation processes aimed at producing organic acids. Although the body of work remains relatively limited, several studies have reported that micro-aeration can enhance organic acid production. Wen et al. (2022) reported that micro-aeration increased the yield of VFAs from activated sludge fermentation by more than twofold and significantly elevated the proportion of acetic acid. Cao et al. (2022b) conducted acidogenic fermentation using agricultural waste as the substrate and found that micro-aeration inhibited the conversion of lactic acid (LA) into VFAs, thereby improving LA accumulation. Similarly, Liao et al. (2025) also confirmed that micro-aeration strategies can promote LA production during the fermentation of herbal waste. These studies indicate that low-level oxygen inputs can be harnessed to steer carbon flux toward specific acidogenic products, including LA.

FW constitutes a major fraction of municipal solid waste and continues to increase globally, posing substantial challenges for urban waste management. AD is widely used for FW treatment due to its operational reliability and energy-recovery capability (Yang et al., 2023); however, it requires long hydraulic retention times (HRTs) and is highly sensitive to environmental disturbances such as the accumulation of free ammonia and VFAs (Prasanna Kumar et al., 2024). In contrast, anaerobic fermentation for LA has recently emerged as a promising strategy to enhance the resource recovery value of FW (Wang et al., 2020). LA has broad applications in the chemical and feed industries and serves as a key monomer for polylactic acid (PLA), a widely used biodegradable polymer (López-Gómez et al., 2019). To date, various strategies—including temperature optimization (Tang et al., 2016), pH control (Hussain et al., 2017), specialized inocula (Arras et al., 2019), and regulation of the organic loading rate (OLR) (Jiang et al., 2013)—have been investigated to improve FW-to-LA conversion. However, existing micro-aeration studies on LA production have focused on agricultural residues or herbal waste, and micro-aerated LA fermentation from FW has not yet been explored. Moreover, the mechanistic basis of micro-aeration in AF systems remains poorly understood, and gene-expression-level insights into community metabolism and functional dynamics are critically needed.

In this study, we investigated the effects of different aeration rates on LA production from FW fermentation, with the aim of determining optimal aeration parameters. Meanwhile, 16S rRNA gene and Internal Transcribed Spacer (ITS) amplicon sequencing were conducted to examine bacterial and fungal community structures under varying aeration conditions. In addition, metatranscriptomic analysis was employed to assess the impact of aeration on microbial gene expression, thereby providing deeper insights into the mechanisms by which micro-aeration enhances LA fermentation. This study not only advances our understanding of micro-aeration as a fermentation strategy but also contributes to the mechanistic foundation and practical optimization of micro-aeration for broader applications in anaerobic bioprocess engineering.

2. Materials and methods

2.1. Substrate and inoculum

To keep the feedstock identical across the whole experimental period, simulated FW was used in the experiment. The full-price commercial dog food (Sam's Club, USA) and polished white rice were mixed at a mass ratio of 1:1, followed by the addition of water for cooking. Subsequently, the mixture was crushed using a wall-breaking machine into a pulp with an average particle size of less than 0.5 mm. The pulp, after homogenization, was stored in a refrigerator at –20°C. Before use, it was thawed, and its total solid content was adjusted to approximately 5.0 ± 0.2 % using tap water as feedstock. The characteristics of the feedstock are shown in Table 1. Biogas slurry used for inoculation was obtained from a FW treatment plant employing mesophilic anaerobic

Table 1
Physicochemical characteristics of FW slurry used as fermentation feedstock.

Item	FW	Biogas slurry
Total solid (TS, %)	5.0 ± 0.2	1.9 ± 0.2
Volatile solid (VS, %)	4.4 ± 0.1	0.9 ± 0.3
Total chemical oxygen demand (TCOD, g/L)	46.8 ± 0.9	7.6 ± 0.4
Soluble chemical oxygen demand (SCOD, g/L)	13.2 ± 0.3	2.0 ± 0.2
pH	6.16 ± 0.32	7.92 ± 0.11
Total carbohydrates (g/L)	30.6 ± 2.4	0.8 ± 0.1
Starch (g/L)	23.1 ± 1.8	0.2 ± 0.1
Cellulose (g/L)	6.2 ± 1.1	0.4 ± 0.1
Crude protein (g/L)	9.3 ± 1.2	3.6 ± 0.2
Crude fat (g/L)	3.5 ± 0.3	0.4 ± 0.1

digestion as its core process, with a treatment capacity of 500 t/d (Zhao et al., 2024a).

2.2. Reactor set-up and operation

Fermentation experiments were conducted in four glass reactors, each with a working volume of 1,500 mL. Each glass reactor was fitted with two 3 mm-diameter ports at the top, functioning as the air inlet and outlet, respectively. The outer side of the air inlet port was connected to a peristaltic pump (LongerPump BT100-3 J, Longer Pump Co., Ltd., China) to supply air into the reactor, while the inner side was connected to a circular high-pressure microporous ceramic diffuser (QL113-0, 130 × 130 mm; Qili Environmental Protection, China). It dispersed the incoming air into fine bubbles with an average diameter of less than 0.5 mm, as specified by the manufacturer. Four aeration rates (the volumetric airflow supplied per unit time) were applied: 0, 5.0, 15.0, and 40.0 mL/min, corresponding to reactors A0, A5, A15, and A40, respectively. A magnetic stir bar was placed in each reactor, and all reactors were immersed in a thermostatic water bath equipped with magnetic stirring. The stirring speed was maintained at 60 rpm, and the temperature was controlled at 35.0 ± 1.0°C. At the start of the experiment, each reactor was loaded with 1,400 mL of substrate and 100 mL of inoculum, resulting in an effective working volume of 1,500 mL. To investigate the effect of aeration on rapid LA fermentation with a high dilution rate, the reactors were operated in a semi-continuous mode with a HRT of 3 days. Each day, 500 mL of fermentation broth was withdrawn and replaced with fresh feedstock, corresponding to an OLR of 14.7 g VS/(L·d). The fermentation experiment lasted for a total of 22 days. Effluent samples were collected every two days for subsequent analysis.

2.3. Fermentation degree

Fermentation degrees can be expressed using hydrolysis rate (HR) and acidogenic rate (AR). These parameters reflect the extent to which solid substrates are solubilized and further converted into fermentation products. They were calculated as follows:

$$HR = (SCOD/TCOD) \times 100\% \quad (1)$$

$$AR = (CODp/TCOD) \times 100\% \quad (2)$$

where SCOD is the soluble chemical oxygen demand of fermentation effluent, TCOD is the total chemical oxygen demand of feedstock, and CODp is the sum of COD for the fermentation products.

2.4. Microbial analysis

On the final day of the experiment (steady-state), fermentation broth samples were collected, immediately frozen in liquid nitrogen, and stored in a liquid nitrogen tank to preserve sample integrity. The frozen samples were then shipped to Majorbio Inc. (Shanghai, China) for DNA/RNA extraction and sequencing.

To analyze microbial diversity, total DNA was extracted using the PowerSoil® DNA Isolation Kit (Mo Bio Laboratories Inc., Carlsbad, USA) and amplified with a PCR thermocycler (GeneAmp® 9700, ABI, USA). For bacteria, the primer pair 338F (5′-ACTCCTACGGGAGGCAGCAG-3′) and 806R (5′-GGACTACHVGGGTWTCTAAT-3′) was used, while for eukaryotes such as fungi, ITS1F (5′-CTTGGTCATTTAGAGGAAGTAA-3′) and ITS2R (5′-GCTGCGTTCTTCATCGATGC-3′) were used. PCR products were purified and sequenced on an Illumina MiSeq platform (Illumina, USA). Sequence data were processed and analyzed using the QIIME 2 software package.

For metatranscriptomic analysis, total RNA was extracted and purified using the Soil RNA Extraction Kit (Majorbio, Shanghai, China) following the manufacturer's instructions. RNA concentration and integrity were assessed using a NanoDrop 2000 spectrophotometer (Thermo Scientific, MA, USA) and an Agilent 5300 Bioanalyzer (Agilent Technologies, Palo Alto, CA, USA). RNA libraries were prepared using the Illumina® Stranded mRNA Prep Ligation Kit (Illumina, San Diego, CA, USA), and ribosomal RNA was removed using the RiboCop rRNA Depletion Kit for Mixed Bacterial Samples (Lexogen, USA). Paired-end (150 bp) sequencing was performed on the Illumina NovaSeq 6000 platform at Majorbio Bio-Pharm Technology Co., Ltd. (Shanghai, China).

Raw sequencing reads were quality-filtered using fastp (v0.24.0) to remove adapter sequences, low-quality reads (threshold: −q 30, −u 20), reads containing ambiguous bases (N > 5), and PCR duplicates, yielding high-quality clean reads for downstream analysis (Chen, 2023). Over-represented rRNA fragments were removed using SortMeRNA (v4.3.6) (Evguenia et al., 2012). De novo assembly was conducted using MEGAHIT (v1.2.9) with optimized k-mer settings (k-min = 21, k-max = 141) (Li et al., 2016). Contigs shorter than 300 bp were discarded. All contigs ≥ 300 bp across samples were pooled and subjected to open reading frame (ORF) prediction using Prodigal (v2.6.3), and ORFs shorter than 300 bp were removed (Hyatt et al., 2021). Redundant sequences were clustered using CD-HIT (v4.8.1) at 97 % nucleotide identity and 90 % coverage, retaining the longest sequence in each cluster to generate a non-redundant gene catalog (Li and Godzik, 2006), with the longest sequence in each cluster retained as the representative to construct a non-redundant gene catalog. Functional annotation was performed using KofamKOALA (Kanehisa et al., 2016). To quantify gene expression and obtain microbial community profiles, paired-end reads were aligned to the non-redundant gene catalog using Salmon (v1.10.3) (Patro et al., 2017), and transcript abundance was reported in transcripts per million (TPM).

2.5. Analytical procedures

pH, oxidation–reduction potential (ORP), and dissolved oxygen (DO) were measured using a digital meter (PHS–3C, INESA, China) fitted with corresponding probes: a pH probe (Model 501), an ORP probe (Model 501), and a DO probe (DO-958-S), all supplied by INESA, China. Due to the high solids content in the effluent, liquid samples were first centrifuged at 6000 × g for 5 min, and the supernatant was filtered through a 0.45 μm pore-size membrane. The resulting filtrate was used for the determination of soluble parameters, including SCOD and fermentation products. TS and VS were measured by the gravimetric method, involving drying at 105 °C and ignition at 600 °C. Total carbohydrate content was determined using the phenol–sulfuric acid method with glucose as the standard. Crude protein was quantified by the Kjeldahl method using Kjeldahl digestion vials (TNT880-CN, Hach, USA) and a DR3900 spectrophotometer (Hach, USA). Crude fat content was determined via the Soxhlet extraction method. Ethanol and VFAs, including acetic acid, propionic acid, *n*-butyric acid, *n*-valeric acid, and *n*-caproic acid, were analyzed using a gas chromatograph (GC-2014, Shimadzu, Japan) equipped with a capillary column (DB-FFAP, 30 m × 0.32 mm × 1.00 μm) and a flame ionization detector (FID). LA concentrations were determined by high-performance liquid chromatography (LC-2030C 3D Plus, Shimadzu, Japan) using an InertSustain® C18 column (5 μm, 25 cm × 4.6 mm) with ultraviolet detection at 206 nm. The mobile phase consisted of 85 % 20 mmol/L KH_2PO_4 and 15 % methanol, delivered at a flow rate of 0.7 mL/min. Soluble glucose was not monitored as a separate parameter because, under the operating conditions of this study, readily fermentable hexoses were expected to be rapidly consumed and therefore unlikely to accumulate.

2.6. Statistical analysis and figure making

Data collected during the steady-state period—defined as the stage in which the concentrations of main fermentation products exhibited relative standard deviations (RSD) of less than 20 %—were used for

subsequent analysis and comparison. Results are presented as mean ± standard deviation and represent the fermentation performance under each corresponding condition. Statistical analyses were performed using R (version 4.3.2; RStudio version 2024.12.1) and Microsoft Excel. Differences in means between two groups were assessed using independent samples *t*-tests, with statistical significance defined as $p \leq 0.05$. Figures were generated using OriginPro 2025 and Microsoft PowerPoint.

3. Results and discussion

3.1. Fermentation performance under different micro-aerobic conditions

3.1.1. Temporal dynamics of fermentation products under different aeration levels

All four fermentation reactors were operated continuously for 22 days. Reactors A5, A15, and A40 were supplied with continuous aeration throughout the entire operation, while reactor A0 was maintained under strictly anaerobic conditions as a control. Based on substrate characteristics and operating conditions, the corresponding aeration loads of 0, 109, 327, and 872 mL air/(gVS·L), respectively. And the reactors reached steady state on Days 10, 14, 6, and 8, respectively. At steady state, the DO concentrations in all reactors remained below 0.5 mg/L. Notably, the DO levels in A0 and A5 were below the detection limit of the probe (<0.1 mg/L). Temporal variations in key fermentation products and pH throughout the experimental period are presented in Fig. 1a.

Immediately after fermentation was initiated, all reactors exhibited rapid and substantial LA accumulation, accompanied by detectable levels of acetic acid and ethanol. The sharp increase in organic acid production caused a rapid decline in pH, with values in all reactors dropping below 3.4 by day 8. These characteristics indicate that all systems rapidly entered the lactic acid-type fermentation stage following a short adaptation phase (Yang et al., 2021). In the early stage of fermentation, low concentrations of propionic acid and butyric acid were also detected. Notably, butyric acid concentrations gradually

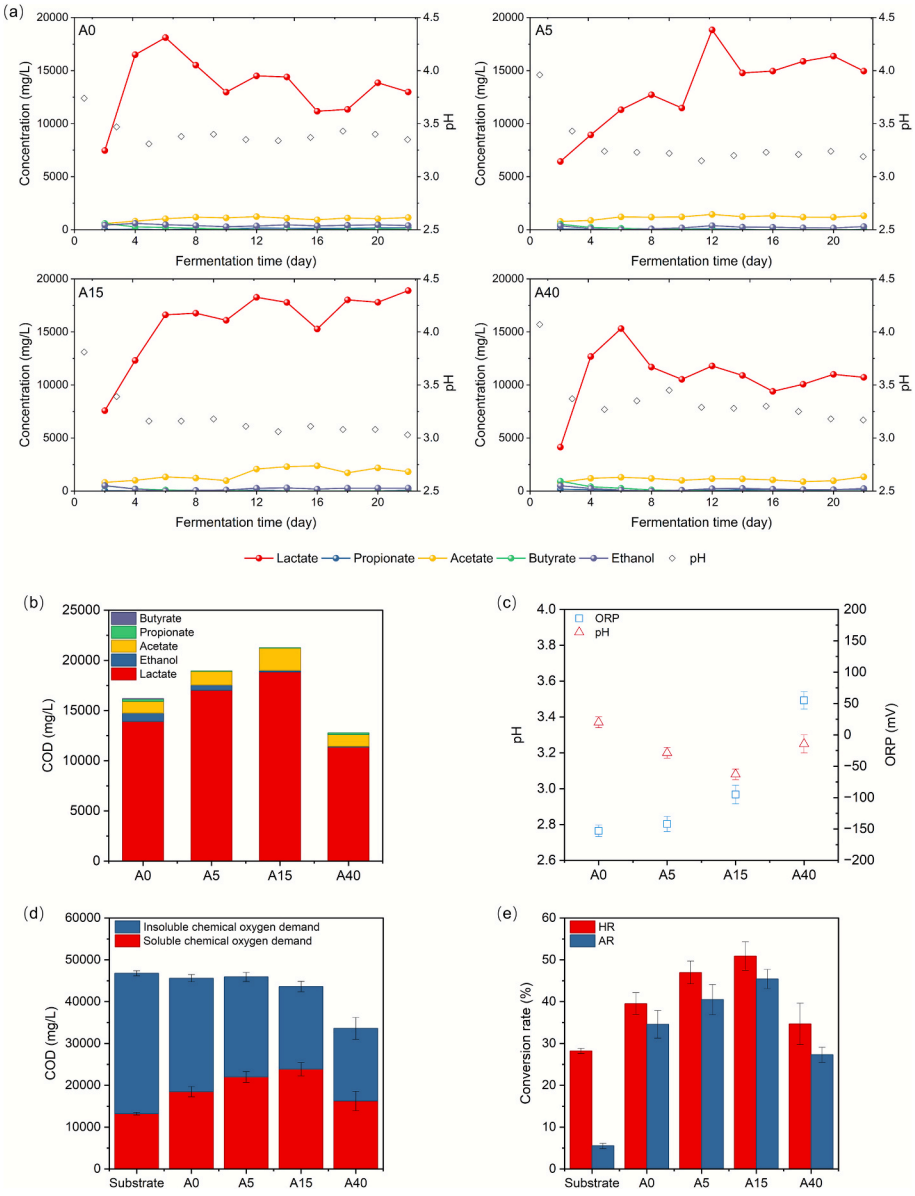


Fig. 1. Fermentation performance of the semi-continuous experiment. (a) Temporal variations in primary fermentation products over the course of the experiment. (b) COD-based distribution of fermentation products. (c) Average pH and ORP values at steady state in different reactors. (d) Distribution of soluble and insoluble COD at steady state across reactors. (e) Hydrolysis and acidification rates under different fermentation conditions at steady state.

declined from initial values of 489–931 mg/L to nearly zero as the reaction progressed. This phenomenon may be attributed to the poor acid tolerance of butyrate-producing bacteria such as *Clostridium* spp. (Esquivel-Elizondo et al., 2017), which were likely suppressed or outcompeted as pH declined.

Despite operating under strict anaerobic conditions, reactor A0 demonstrated the fastest initial LA accumulation, reaching a peak concentration of 18,116 mg/L by day 6. However, the sharp pH drop (to 3.31 on day 6) likely inhibited the activity of lactic acid bacteria (LAB), leading to a gradual decrease in LA concentration. The system eventually reached a steady state with a stabilized pH of 3.37 ± 0.03 and LA concentration maintained at 13,049 ± 1,446 mg/L. In contrast, reactors A5 and A15, which received low to moderate aeration, exhibited slower initial LA accumulation. Nevertheless, excluding short-term system fluctuations, both reactors showed an overall upward trend in LA concentration before reaching steady state. Correspondingly, their steady-state pH values (3.20 ± 0.03 and 3.08 ± 0.03, respectively) were lower than that observed in reactor A0. These observations suggest that low-level aeration may not accelerate the proliferation of LAB and facilitate rapid system start-up, but could enhance the microbial community's tolerance to acidic conditions. This effect may be mediated by the induction of acid-resistance mechanisms within existing microorganisms or by reshaping the community toward more acid-tolerant taxa. At the highest aeration rate, reactor A40 exhibited a similar trend to A0, with LA concentration initially increasing and then declining. However, unlike A0, the subsequent decrease in LA levels in A40 was likely attributable to the aerobic consumption of organic acids by oxygen-respiring microorganisms (Cao et al., 2022a). Consequently, the system stabilized at a higher pH of 3.25 ± 0.05 compared to A5 and A15.

3.1.2. Steady-state product distribution, COD balance, and substrate conversion efficiency

Fig. 1b summarizes the COD-based distribution of fermentation products across reactors during the steady-state phase. LA was the predominant product in all systems, accounting for more than 85 % of the total product COD. Low-level aeration effectively enhanced LA production. At an aeration rate of 109 mL air/(gVS·L), reactor A5 reached a steady-state LA concentration of 15,965 ± 1,405 mg/L. Further increasing the aeration rate to 327 mL air/(g VS·L) in reactor A15 resulted in the highest LA concentration observed in this study (17,676 ± 1,135 mg/L), corresponding to an LA yield of approximately 0.40 g LA/g VS—representing a 35.5 % increase compared to the strictly anaerobic reactor A0. Under excessive aeration (A40), the LA concentration dropped significantly to 10,647 ± 754 mg/L, and the acetic acid concentration was also lower than that observed in A15. In addition to LA, acetic acid and ethanol were also important fermentation products. Under low aeration conditions (≤ 327 mL air/(gVS·L)), the concentration of acetic acid increased with rising aeration rates, whereas ethanol exhibited an opposite trend.

The introduction of air into the fermentation system enabled partial microbial oxidation of organic matter, leading to a decrease in the TCOD of the fermentation broth relative to the feedstock—a phenomenon particularly evident in reactor A40 (Fig. 1d). Despite consistent feedstock properties throughout the experiment, A40 exhibited a significant reduction in TCOD, with values approximately 26.3 % lower than those of A0 ($p < 0.001$). In contrast, TCOD changes in A5 and A15 were not statistically significant, likely due to their lower aeration rates.

Fig. 1e shows the HR and AR of each system during the steady-state phase. Before fermentation, most of the organic matter in the feedstock remained in the solid phase, as indicated by a low HR of 28.2 ± 0.6 %. Under strictly anaerobic conditions (A0), hydrolytic microorganisms facilitated the breakdown of solid substrates into soluble compounds, resulting in an average HR of 39.5 % and an AR of 34.6 %. Most of the solubilized organic matter was subsequently converted into fermentation products. Trace levels of aeration further enhanced substrate solubilization, with reactors A5 and A15 achieving HRs of 47.0 % and 50.9 %, respectively. The improved hydrolysis also supported more effective acidification, with A15 reaching the highest AR of 45.4 %. These findings are consistent with previous studies showing that low-level aeration can stimulate the production of extracellular hydrolytic enzymes and enhance acidification efficiency (Cao et al., 2022b). However, when the aeration rate was further increased (A40), the rate of aerobic consumption of organics outpaced the benefits of enhanced hydrolysis, leading to a marked decline in both HR and AR.

3.2. Microbial community

Based on metatranscriptomic data from steady-state samples, domain-level activity profiles of microbial communities under different aeration conditions were obtained (Fig. 2a). In previous studies, Bacteria have been recognized as the principal contributors to anaerobic fermentation, while Eukaryota are generally scarce in acidogenic systems (Zhao et al., 2024b). In line with previous findings, Bacteria emerged as the most transcriptionally active domain, with relative expression levels exceeding 90 % across all samples in current study. In A0, Bacteria accounted for approximately 99.1 % of the total transcripts, while Eukaryota represented only 0.6 %. Notably, the transcriptional activity of Eukaryota increased with aeration, reaching 1.0 %, 2.6 %, and 6.4 % in reactors A5, A15, and A40, respectively. This may be explained by the ability of these eukaryotic microorganisms to perform oxidative phosphorylation in mitochondria under micro-aerobic conditions, enabling more efficient energy production and thereby granting them a greater competitive advantage (Zimorski et al., 2019).

16S rRNA gene and ITS amplicon sequencing were performed to further resolve the bacterial and fungal community compositions, and the results are presented in Fig. 2b and Fig. 2c, respectively. Bacterial community profiling indicated that, across all samples, members of the family Lactobacillaceae—classified as LAB—predominated and served as the key functional guild responsible for LA production in this study. In A0, *Lactobacillus amylolyticus* was dominant (51.7 %), followed by *Limosilactobacillus panis*, *Lactobacillus amylovorus*, and *Limosilactobacillus* sp.; together these four LAB accounted for ~ 94.2 % of the bacterial community. Micro-aeration reshaped the community: in A5, relative to A0, *L. amylovorus* increased and became the dominant species (40.5 %), while the other three aforementioned taxa declined. Further increases in aeration reduced community evenness. In A15—the reactor with the strongest LA performance—*L. amylolyticus* achieved overwhelming dominance (86.5 %), followed by *Limosilactobacillus pontis* (5.7 %). Under the highest aeration level (A40), the bacterial community became further simplified, with *L. amylolyticus* and *L. panis* together comprising over 98 % of the total population, while all other taxa individually accounted for less than 0.5 %. *L. amylolyticus* is commonly isolated from strongly acidic, carbohydrate-rich fermentation environments (e.g., acidified malt/wort and tofu whey), and certain strains (e.g., *L. amylolyticus* L6) can maintain high viability and LA production at approximately pH 3 (Fei et al., 2017). Therefore, the increased abundance of *L. amylolyticus* under micro-aeration likely enhanced the overall acid tolerance of the microbial community.

As shown in Fig. 2c, although ITS amplicon sequencing provided limited species-level resolution for fungi, it still offered a broad overview of the fungal community structure. In reactor A0, the community was relatively simple and dominated by *Hyphopichia* sp. (82.4 %), a member of the order Saccharomycetales known for performing ethanol fermentation under anaerobic or hypoxic conditions (Garcia et al., 2021). In A5, increased aeration led to greater fungal diversity, characterized by a decline in *Hyphopichia* sp. and an expansion of other taxa, including uncultured Saccharomycetales and *Hypomyces semicircularis*. Notably, in A15, *Holtermannia saccardoi* proliferated to 17.7 % of the fungal community. This species belongs to the order Holtermanniales, members of which have been reported to rely primarily on aerobic respiration and to assimilate a broad spectrum of carbon substrates (Wuczkowski et al., 2011). Taken together with the domain-level Eukaryota expression

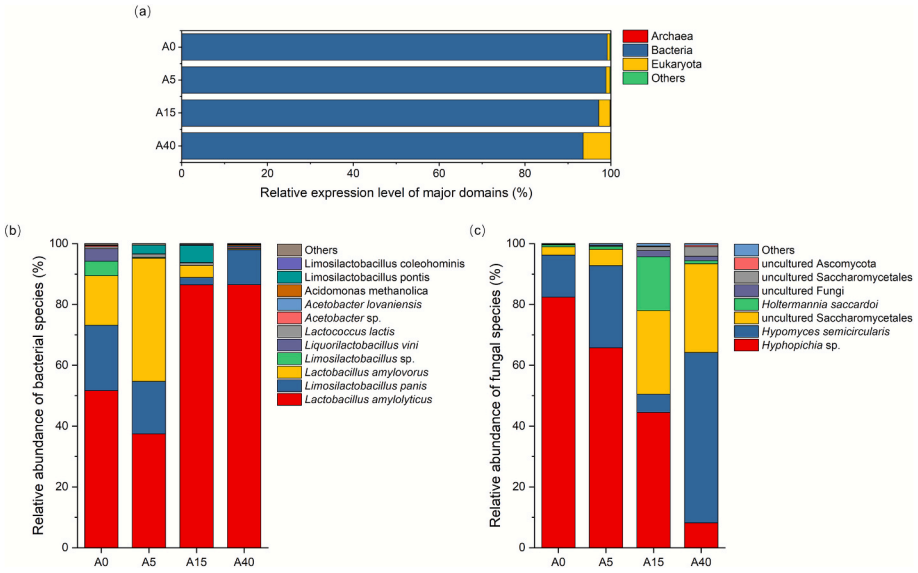


Fig. 2. Microbial community composition under different fermentation conditions during steady state. (a) Domain-level activity profiles of microbial communities under different aeration conditions, based on metatranscriptomic data. (b) Bacterial and archaeal community composition based on 16S rRNA amplicon sequencing. (c) Fungal community composition based on ITS amplicon sequencing.

profiles (Fig. 2a), these results suggest that micro-aeration not only increased fungal community complexity but also enhanced their functional roles in the fermentation system.

3.3. Metatranscriptomic analysis

Through the semi-continuous micro-aeration fermentation experiment, we identified the reactor A15—operated at an aeration rate of 327 mL air/(gVS·L)—as achieving the highest LA yield among all tested conditions. This aeration level was therefore considered optimal for LA production from FW. To elucidate how oxygen input facilitates substrate hydrolysis, improves LA yield, and activates fungal metabolism, we performed an in-depth metatranscriptomic analysis of microbial communities from the representative groups A0 (strictly anaerobic) and A15 (optimal micro-aerated).

3.3.1. Mechanistic insights into microaeration-enhanced hydrolysis

Carbohydrates are widely recognized as the principal substrates metabolized by microorganisms during fermentation processes (Wang et al., 2020). In FW, starch and cellulose are the main carbohydrates (Table 1), composed of glucose units linked by glycosidic bonds (Yang et al., 2023). During fermentation, microbial extracellular enzymes hydrolyze them into oligosaccharides and monosaccharides, which then enter carbon metabolism pathways to form fermentation products. To investigate the impact of micro-aeration on carbohydrate degradation, the metabolic pathways for starch and cellulose hydrolysis were summarized based on the KEGG database, and the transcriptional abundance of the corresponding hydrolytic enzymes was analyzed (Fig. 3a). In the diagram, solid black arrows denote community-level metabolic routes with strong transcriptional support, whereas dashed arrows represent KEGG-inferred pathways lacking detectable or appreciable transcript levels.

As shown in the figure, in both A0 and A15, starch degradation appears to start with the action of glucan 1,4-α-maltohydrolase (EC 3.2.1.133), which sequentially cleaves maltose units from the non-reducing ends of the starch polymer. The resulting maltose is subsequently hydrolyzed into glucose by maltase (EC 3.2.1.20). In parallel, dextrins, which may be generated during partial starch breakdown, are further hydrolyzed by oligo-1,6-glucosidase (EC 3.2.1.10) to release glucose. Under optimal micro-aeration (A15), the expression levels of these three hydrolases were markedly elevated compared to A0, by 8.8-fold, 2.7-fold, and 5.9-fold, respectively. This clearly demonstrates that micro-aeration strongly stimulated the expression of community-level hydrolase genes, thereby enhancing the HR. In contrast, cellulolytic enzyme expression remained low across both conditions, suggesting minimal cellulose degradation. This observation aligns with previous findings that efficient cellulose utilization typically requires long HRT, and is rarely achieved in short-HRT acidogenic fermentation systems (Yang et al., 2023).

To further identify the microbial contributors responsible for starch hydrolysis, we analyzed the taxonomic origins of transcripts encoding these three key enzymes (Fig. 3b). *L. amylolyticus* was the predominant contributor to their expression. In A15, this species exhibited a marked increase in enzyme transcript abundance compared to A0, while other microbial taxa showed reduced expression. Consequently, the proportion of transcripts derived from *L. amylolyticus* exceeded 95 % of the total reads assigned to these enzymes. These results suggest that micro-aeration not only promoted the proliferation of *L. amylolyticus* (as previously described in Section 3.2), but also enhanced its hydrolase secretion capacity. Together, these effects likely improved substrate availability and facilitated downstream LA production.

3.3.2. Mechanistic insights into microaeration-enhanced lactic acid production

Glucose released from starch hydrolysis serves as the primary substrate for LA fermentation. It is metabolized by LAB mainly through three distinct pathways: homolactic fermentation, heterolactic fermentation, and the Bifidobacteria pathway. Due to the low abundance of Bifidobacteria and the negligible transcriptional activity of their key enzymes revealed by metatranscriptomic analysis, this chapter focuses on analyzing and comparing the expression profiles of the homolactic and heterolactic fermentation pathways between A0 and A15. In addition, the mechanism by which the microbial community utilizes exogenous oxygen is also explored.

Both homolactic and heterolactic fermentation pathways utilize glyceraldehyde-3-phosphate (G3P) as a key intermediate, which is subsequently reduced to LA via pyruvate. The main distinction lies in how G3P is generated: homolactic fermentation employs the glycolytic pathway (Embden–Meyerhof–Parnas, EMP), while heterolactic fermentation relies on the phosphoketolase (PK) pathway (Wang et al.,

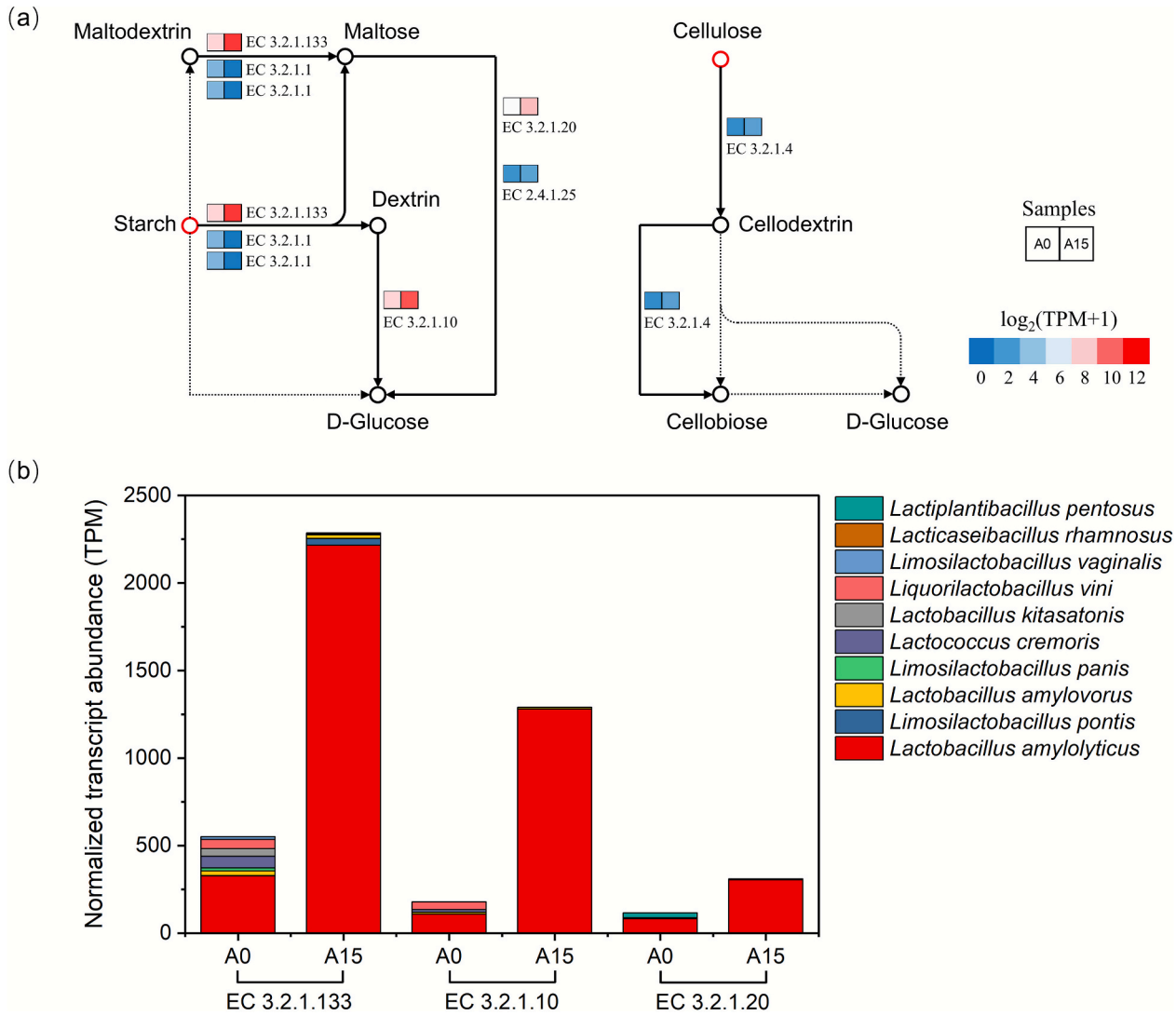


Fig. 3. Metabolic pathways of starch and cellulose hydrolysis and transcriptional expression levels of key enzymes under different fermentation conditions. (a) Schematic overview of the metabolic pathways involved in starch and cellulose hydrolysis, along with the transcriptional expression levels of key enzymes in samples A0 and A15. (b) Transcript quantification and taxonomic origins of genes encoding key enzymes responsible for starch hydrolysis, including glucan 1,4-α-maltohydrolase (EC 3.2.1.133), oligo-1,6-glucosidase (EC 3.2.1.10), and maltase (EC 3.2.1.20).

2021). As shown in Fig. 4a, in A0, the transcriptional levels of most key enzymes in the EMP pathway were higher than those in the PK pathway (except phosphogluconate dehydrogenase, EC 1.1.1.343), suggesting that homolactic fermentation predominated under anaerobic conditions. In the micro-aerated system (A15), EMP pathway activity was further enhanced, as evidenced by the upregulation of related enzyme transcripts. Notably, transcriptional levels of pyruvate oxidase (EC 1.2.3.3) and acetate kinase (EC 2.7.2.1) increased by 7.2-fold and 1.5-fold in A15, respectively, compared to A0. Pyruvate oxidase catalyzes the conversion of pyruvate to acetyl-P using O_2, generating H_2O_2 as a byproduct. Acetyl-P can then participate in substrate-level phosphorylation mediated by acetate kinase, producing ATP and acetic acid. Simultaneously, NADH peroxidase (EC 1.11.1.1)—which detoxifies H_2O_2 by converting it to H_2O while consuming NADH—was upregulated by 3.0-fold in A15. Since NADH is typically consumed during the reduction of pyruvate to LA in conventional homolactic fermentation, this suggests that under micro-aerobic conditions, homofermentative LAB may reroute metabolic flux toward acetic acid production via the acetyl-phosphate pathway, utilizing NADH for the detoxification of reactive oxygen species such as H_2O_2.

Maintaining efficient ATP generation and appropriate NADH turnover is essential for fermentation performance and microbial homeostasis. While ATP provides energy for cellular activities, the recycling of NADH is essential for maintaining metabolic flux(de Graef et al., 1999). Under micro-aerobic conditions, the incorporation of oxygen-responsive alternative pathways not only enhances ATP yield but also promotes NADH reoxidation, thereby maintaining intracellular redox balance. Specifically, the alternative shunted acetate kinase pathway converts acetyl phosphate to acetate with concomitant ATP production, while NADH peroxidase catalyzes the reduction of H_2O_2, thereby regenerating NAD^+. Similar oxygen-regulated metabolic mechanisms have been reported previously (Zotta et al., 2017). The following equations illustrate the theoretical energy yields and redox balances under different fermentation conditions, including anaerobic LA production via the EMP pathway (Eq. (1)), partial pyruvate oxidation under micro-aerobic conditions (Eq. (2)), and complete pyruvate oxidation under micro-aerobic conditions (Eq. (3)):

$$C_6H_{12}O_6 + 2ADP \rightarrow 2C_3H_6O_3 + 2ATP \quad (1)$$

$$C_6H_{12}O_6 + O_2 + 3ADP \rightarrow C_3H_6O_3 + C_2H_4O_2 + 3ATP + H_2O + CO_2 \quad (2)$$

$$C_6H_{12}O_6 + 2O_2 + 4ADP \rightarrow 2C_2H_4O_2 + 4ATP + 2H_2O + 2CO_2 \quad (3)$$

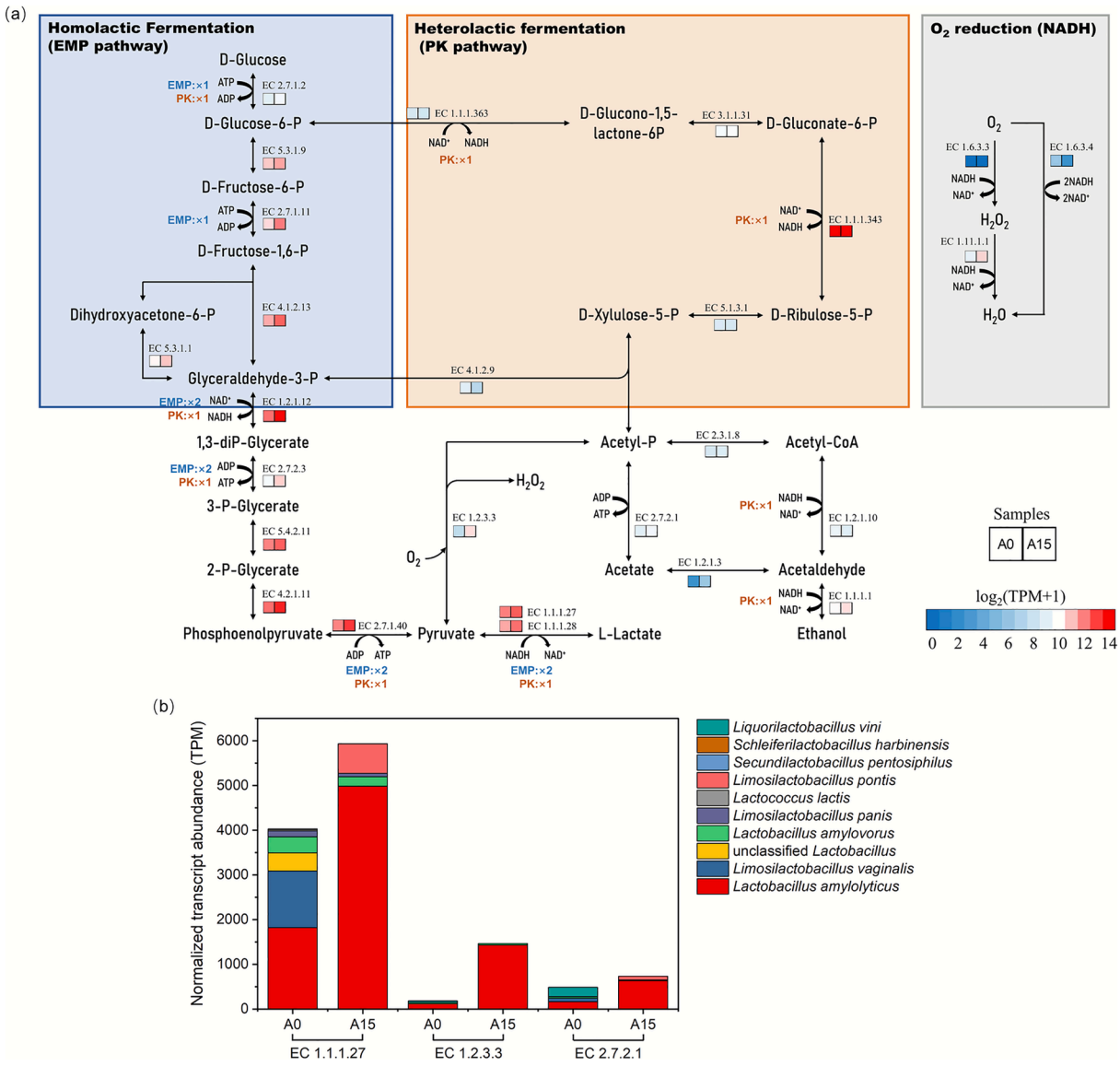


Fig. 4. LA production pathways, oxygen-responsive metabolic routes, and transcriptional profiles of the microbial community in A0 and A15. (a) Schematic representation of homolactic and heterolactic fermentation pathways, as well as oxygen-utilization pathways, with key enzymes annotated according to the KEGG database and overlaid with transcriptional expression levels. (b) Transcript quantification and taxonomic origins of genes encoding key enzymes involved in the oxygen response of homolactic LAB, including L-lactate dehydrogenase (EC 1.1.1.27), pyruvate oxidase (EC 1.2.3.3), and acetate kinase (EC 2.7.2.1).

The actual metabolic reactions occurring in A15 are likely a composite of these three routes, proceeding simultaneously in specific stoichiometric proportions.

Pyruvate oxidase and acetate kinase represent key enzymatic responses of homofermentative LAB to exogenous oxygen. *L. amylolyticus*, a prototypical obligate homofermentative species (Felis and Dellaglio, 2015), was identified as the primary contributor to the transcriptional activity of both enzymes based on metatranscriptomic data (Fig. 4b). This indicates that *L. amylolyticus* effectively harnesses the energetic benefits of oxygen-driven pyruvate oxidation to acetate. The increased ATP yield from this pathway likely supports its robust proliferation and ecological dominance under micro-aerated conditions (Fig. 2b). Consequently, *L. amylolyticus* emerged as the dominant LA producer in A15, accounting for 84 % of the total transcriptional abundance of L-lactate dehydrogenase (EC 1.1.1.27), compared to only 45 % in A0. This also explains the higher acetate production observed in A15 relative to A0.

Heterofermentative LAB are generally known to possess a complete enzymatic toolkit for coping with exogenous oxygen (Condon, 1987). Under anaerobic conditions, NADH generated via the PK pathway is typically used to reduce acetyl-CoA, resulting in ethanol production. Accordingly, the heterofermentative fermentation of 1 mol of glucose yields one mole of LA and 1 mol of ethanol. In the presence of oxygen, these organisms are capable of reoxidizing NADH via NADH oxidases, such as NADH:O_2 oxidase (EC 1.6.3.4) or NADH:H_2O_2 oxidase (EC 1.6.3.3), thereby shifting the metabolic flux toward acetate formation via the acetate kinase pathway and enabling additional ATP generation (Condon, 1987).

However, in A15, the expression levels of both NADH oxidases were extremely low, and most key enzymes associated with the PK pathway also exhibited minimal transcriptional activity. Concurrently, the contribution of heterofermentative LAB—including *L. pontis*, *Limosilactobacillus vaginalis*, and *L. panis* (Felis and Dellaglio, 2015)—to the total transcripts encoding L-lactate dehydrogenase (EC 1.1.1.27) was lower in A15 than in A0. These findings indicate that micro-aeration did not promote the proliferation of heterofermentative LAB as might be expected based on conventional theories. One plausible explanation is that homofermentative LAB possess a higher affinity for oxygen,

enabling them to rapidly scavenge and deplete the limited O_2 supply under micro-aerobic conditions. As a result, oxygen becomes unavailable to heterofermentative counterparts before they can initiate their own oxidative metabolism. This early and preferential oxygen consumption allows homofermentative LAB to gain a metabolic advantage, leading to their rapid proliferation and dominance in LA production. In contrast, heterofermentative LAB are effectively excluded from accessing the oxidative energy pathways and gradually lose their ecological niche within the community. This explanation is also consistent with the lower ethanol concentrations observed in A15 compared to A0.

3.3.3. *Mechanistic insights into the stimulation of eukaryotic activity by microaeration*

Compared with strictly anaerobic conditions, prokaryotic microorganisms played a more prominent role in driving community function under micro-aerobic conditions (Fig. 2a). Metatranscriptomic analysis revealed that the transcriptional abundance of genes involved in oxidative phosphorylation (map00190, KEGG) in sample A15 was approximately 30 % higher than that in A0. Furthermore, cDNA read origin analysis indicated that transcripts derived from eukaryotes increased by approximately 3.8-fold in A15 compared to A0 (Fig. 5a). Oxidative phosphorylation—the electron transport chain (Complexes I–IV) coupled to ATP synthase (Complex V)—occurs on the inner mitochondrial membrane. Electrons from NADH (via Complex I) and $FADH_2$ (via Complex II) flow to ubiquinone, then Complex III, cytochrome *c*, and Complex IV, which reduces O_2 to H_2O. Complexes I, III, and IV pump protons from the matrix to the intermembrane space, generating a proton-motive force that drives ATP synthesis by Complex V (Nolfi-Donegan et al., 2020). In A15, the gene expression levels of representative subunits of the respiratory complexes—including *ND1–ND6* (Complex I), *SdhA* and *SdhB* (Complex II), *UQCRFS1* and *CYC1* (Complex III), *COX1*, *COX4*, *COX5*, and *COX6* (Complex IV), as well as *CYC* (cytochrome *c*)—were significantly upregulated (see Supplementary Material). These results collectively indicate a substantial enhancement in the activity of the complete eukaryotic respiratory chain under micro-aerobic conditions in A15. Meanwhile, the transcriptional abundance of eukaryote-derived genes associated with the citrate cycle (TCA cycle; map00020, KEGG) in A15 was approximately 19.6-fold higher than that in A0 (Fig. 5b). As the central hub of aerobic carbon metabolism, the activation of the TCA cycle supports the notion that eukaryotes decompose organic substrates via aerobic pathways.

To further elucidate the carbon source utilization strategies of eukaryotes, we analyzed the expression of gene *HK*, *lldD*, and *acd*, which encode key enzymes responsible for the initial assimilation of hexoses, LA, and acetic acid, respectively (Fig. 5c–e). All three enzymes were found to be of fungal origin, and their expression levels in A15 were significantly higher than those in A0. These results suggest that fungi exhibit enhanced uptake and assimilation of these carbon substrates under micro-aerobic conditions.

Taken together, the evidence indicates that under micro-aerobic conditions, eukaryotes—particularly fungi—assimilate glucose and fermentation metabolites such as LA and acetic acid, and oxidize them via the TCA cycle through a fully electron transport chain (ETC). When the aeration rate was further increased (A40), a substantial portion of

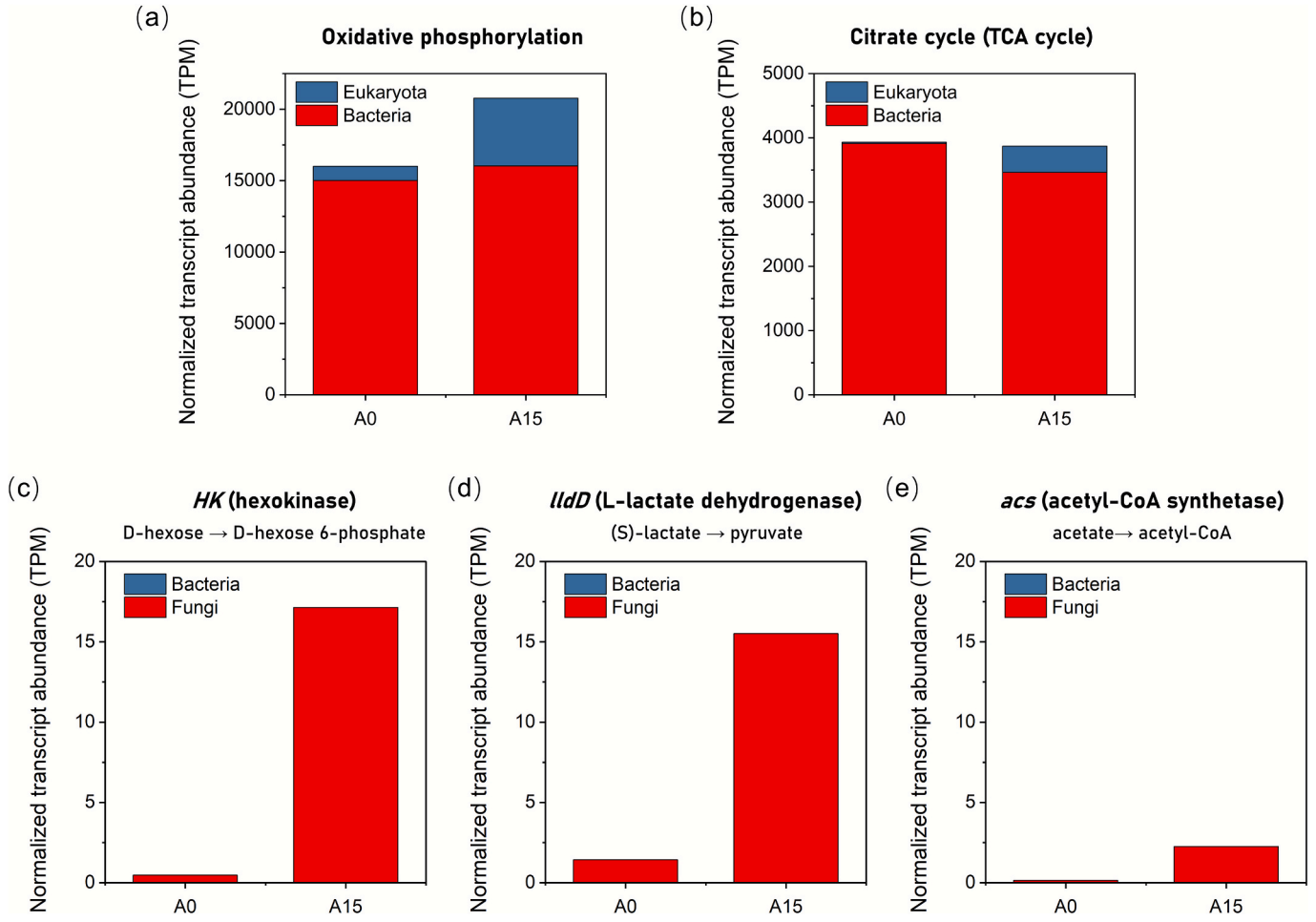


Fig. 5. Transcript quantification and taxonomic origins of genes involved in aerobic metabolism and carbon source assimilation in samples A0 and A15. (a) Genes associated with oxidative phosphorylation (map00190, KEGG). (b) Genes related to the citrate cycle (map00020, KEGG). (c–e) Expression levels and taxonomic origins of key enzymes responsible for the initial assimilation of hexoses (c), LA (d), and acetic acid (e), respectively.

organic substrates or intermediate metabolites was likely consumed through aerobic respiration by fungi, which may explain the lower LA concentration as well as the reduced HR and acidification rate AR observed in A40.

3.4. Integrative mechanistic model of micro-aeration effects on lactic acid fermentation

Based on the aforementioned findings, a comprehensive mechanistic model was developed to illustrate how micro-aeration enhances LA fermentation. As shown in Fig. 6, the system-level responses to exogenous oxygen supply are summarized, where red arrows indicate fermentation steps or microbial activities that are promoted under micro-aerobic conditions, and green arrows represent those that are suppressed.

Under anaerobic conditions, LA production is jointly supported by homofermentative and heterofermentative LAB, with eukaryotes being almost entirely absent. Upon micro-aeration, oxygen enters the fermentation broth, triggering specific metabolic responses. For homofermentative LAB, oxygen induces the activation of the pyruvate oxidase–acetate kinase pathway, which diverts part of the pyruvate flux toward acetate formation, accompanied by the production of H_2O_2. This alternative pathway increases ATP yield, thereby supporting the proliferation of homofermentative LAB. Correspondingly, NAD^+ regeneration shifts from the reduction of pyruvate to LA toward the reduction of H_2O_2 via NADH peroxidase. The enhanced energy yield facilitates the rapid proliferation of homofermentative LAB, particularly *L. amylolyticus*, which dominates LA production under micro-aerobic conditions.

Simultaneously, the presence of oxygen enables eukaryotic microorganisms—primarily fungi—to actively oxidize and assimilate carbon substrates such as glucose, LA, and acetic acid via the TCA cycle and ETC, resulting in high metabolic activity. However, when the aeration rate is excessive, overgrowth of eukaryotes leads to substantial consumption of organic acid products, which in turn reduces LA yield and suppresses both the HR and AR of the system.

Although heterofermentative LAB possess a complete enzymatic toolkit to cope with exogenous oxygen and are theoretically capable of benefiting under aerobic conditions, they are ultimately at a disadvantage. This is primarily due to their lower oxygen affinity compared to homofermentative LAB and fungi. Under micro-aerobic conditions, oxygen is rapidly consumed by homofermentative LAB and fungi, leaving little or no oxygen available for heterofermentative LAB to activate their oxygen-dependent pathways. As a result, heterofermentative LAB exhibit reduced abundance and metabolic activity, ultimately losing their competitiveness in the micro-aerated fermentation system.

3.5. Research prospects and limitations

This study demonstrates that moderate micro-aeration can substantially enhance LA fermentation from FW, leading to marked increases in both hydrolysis and acidification efficiencies. These findings are relevant for conventional two-stage anaerobic digestion processes as well as for LA-oriented fermentation systems. From an engineering perspective, micro-aeration could in principle be implemented in existing food-waste anaerobic digestion plants through relatively simple retrofits (e.g., installation of low-rate aeration and basic control), although further validation and scale-up studies are still required.

At the same time, an important limitation of this work should be acknowledged. To ensure a constant feedstock composition throughout the long-term semi-continuous experiments, we used a mixture of commercial dog food and polished rice as a synthetic substrate. Although this substrate contains starch, cellulose, proteins and lipids, which represent the major components of real FW, its detailed composition and physical properties differ from those of real FW streams, whose origins and compositions are much more heterogeneous and time-variable. In practice, real FW may also contain higher lipid contents, more recalcitrant fibrous fractions, and potential inhibitory substances (e.g., salt, seasonings and preservatives), all of which could influence microbial community succession and fermentation

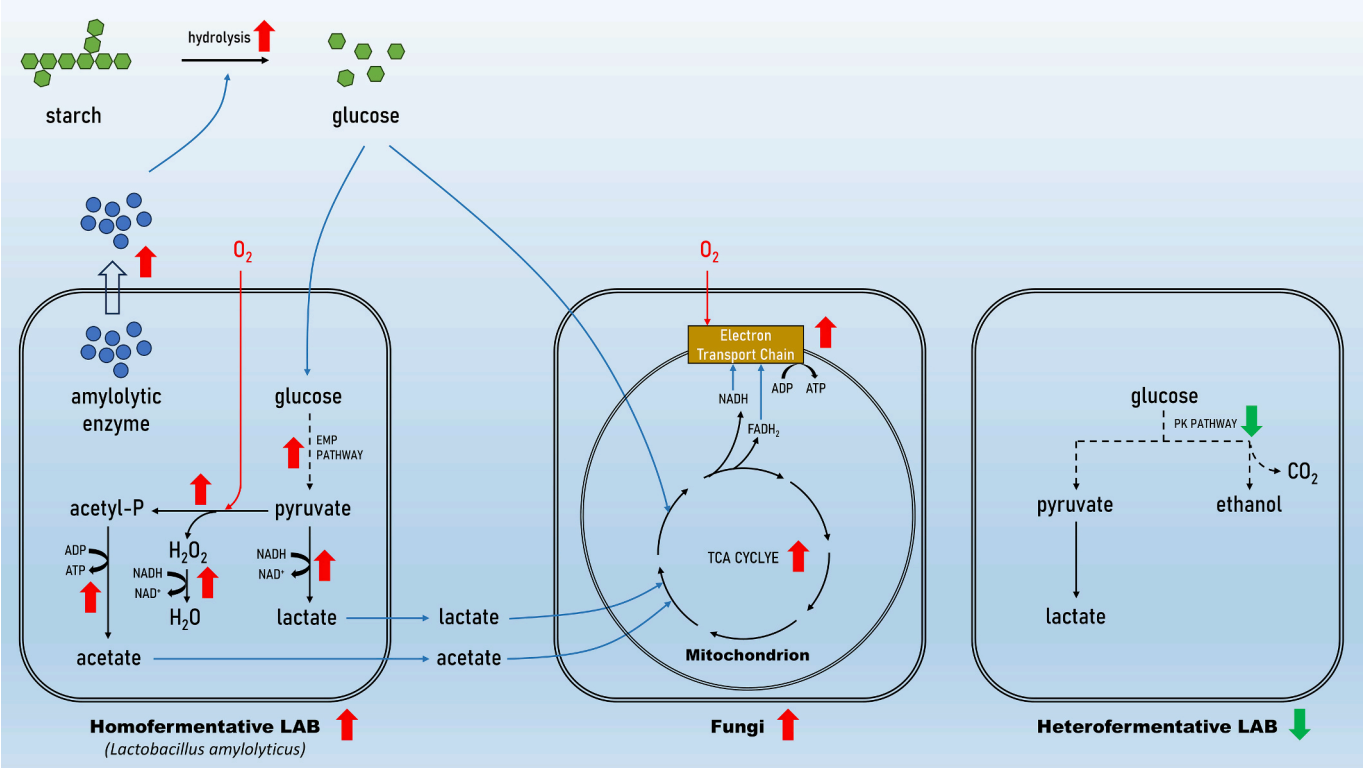


Fig. 6. Mechanistic model of micro-aeration effects on LA fermentation.

performance under micro-aeration. Therefore, future studies using real FW from different sources and with temporally varying compositions are needed to validate the enhancement effect of micro-aeration and the proposed microbial metabolic mechanisms, and to evaluate the robustness and scalability of this strategy under realistic operating conditions.

4. Conclusion

This study investigated the effects of different aeration rates on LA fermentation of FW. Micro-aeration at a rate of 327 mL air/(gVS·L) significantly enhanced LA yield, achieving approximately a 35.5 % increase compared to strictly anaerobic conditions. Amplicon sequencing and metatranscriptomic analyses revealed that micro-aeration activated the pyruvate oxidase–acetate kinase pathway in homofermentative LAB, particularly *L. amylolyticus*. By converting pyruvate to acetate, these microorganisms gained additional ATP, which promoted their proliferation, enhanced the production of starch-degrading enzymes, and ultimately improved the community's LA production efficiency. Moreover, micro-aeration stimulated the proliferation of fungi and enhanced their ability to oxidize organic substrates. In contrast, heterofermentative LAB were outcompeted under micro-aerobic conditions and became marginalized within the fermentation system. E-supplementary data for this work can be found in e-version of this paper online.

CRediT authorship contribution statement

Luxin Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Chuyun Zhao:** Writing – original draft, Investigation, Data curation. **Caiping Sun:** Visualization, Investigation, Data curation. **Huan Li:** Writing – review & editing, Supervision, Resources, Project administration, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2025.133807.

Data availability

The datasets supporting this study—including the physicochemical properties of fermentation substrates and samples, raw 16S/ITS amplicon sequencing data, and raw metatranscriptomic sequencing data—are available from the corresponding author upon reasonable request. Requests must include a brief description of the intended research purpose, and a signed data usage agreement (which will be provided by the authors) is required to ensure compliance with the confidentiality obligations of our collaborating institution.

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